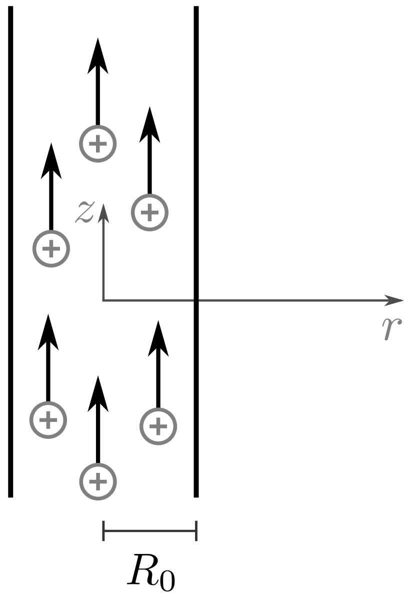
Physik II — FS 2024 — Prof. R. Wallny — Serie 10

Release: 30th April 2024 – **Due date and discussion:** 16th May 2024

Exercise 1: Proton Beam

Objective: Special relativity, Repetition of Gauss' Law, Fields of moving particles.

A particle accelerator produces a proton beam with a kinetic energy of $E_{\text{kin}} = mc^2$ per proton, where m is the proton mass. The beam has a current of I and a radius R_0 in the laboratory frame. Inside the beam, the beam is homogeneous and therefore the current density homogeneous i.e. $\vec{\mathbf{j}}(r) = \text{const.}$ for $r < R_0$.



a) Compute the total energy and the velocity v of the protons in the lab frame.

b) Compute the charge density ρ of the beam in the lab frame as a function of I, R_0 and the velocity of light c.
Hint: The charge is a Lorentz invariant property.

c) Compute the electric field $\vec{\mathbf{E}}$ of the beam in the lab frame for all space.

d) Show that the charge density ρ' in the proton's reference frame is

$$\rho' = \frac{\rho}{\gamma} \tag{1}$$

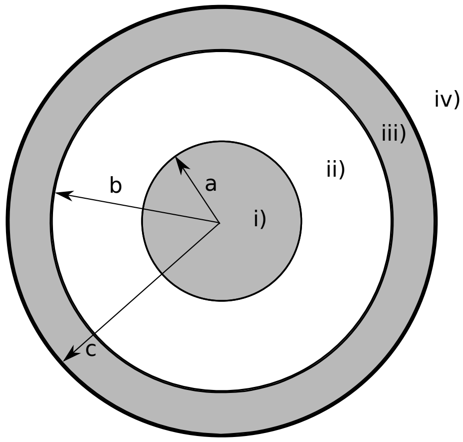
e) Compute the electric field $\vec{\mathbf{E}}'$ in the reference frame of the protons for all space.

f) Due to the electromagnetic interactions between the protons they are slightly steered away from a straight path. Find an expression for the initial acceleration $\vec{\mathbf{a}}(r) = a(r)\vec{\mathbf{e}}_r$ of the protons in the radial direction in the lab frame.

Exercise 2: Ampère's law

Objective: current density, Ampère's law, calculation of line and surface integral.

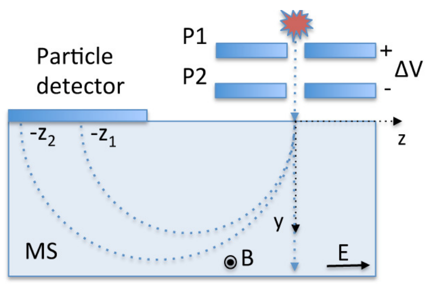
A long cylindrical conductor of radius a is coaxially placed inside an annular cylindrical conductor with inner radius b and outer radius c. The perpendicular cross section is shown in the figure below. The same current I flows through both conductors, but in opposite directions. Find the magnetic field at a point (i) inside the inner conductor, (ii) in the region between the two cylinders, (iii) inside the annular cylinder and (iv) outside both conductors. Assume that the current density is uniform in both conductors.



Exercise 3: Charged particles in electric and magnetic fields.

Objective: Lorentz force, coupled differential equations, motion in electric and magnetic fields.

A particle at rest with positive charge q and mass m is accelerated between two plates P1 and P2 by a potential difference of ΔV. It then enters a region MS with a homogenous electric field of strength E along the z-axis and a homogenous magnetic field of strength B along the x-axis. The x-axis is pointing out of the plane, the y-axis downwards and the z-axis to the right. The particle enters along the y-axis at point (0,0,0) in the lab-system into MS (see figure).



a) Show that the velocity v_0 of the charge particle before its entry into the region MS is $v_0 = \sqrt{2q\Delta V/m}$. Assume that the velocity v_0 is much smaller than the speed of light.

b) Describe the trajectory of the particles in the region MS. To do this, first find the equations of motion and then solve them using the initial conditions. How does the path run in the special case $v_0 = 0$?
Hint: Solve the equations of motions for v and then integrate them to find $\vec{r}(t)$.

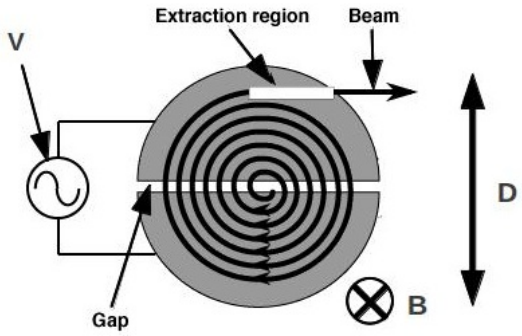
c) Consider the case where strengths of E and B-fields have been adjusted so that the trajectory of the particle in MS region is a straight line along the y-axis. Determine the specific charge q/m of this particle as a function of field strengths E and B and the acceleration voltage ΔV.

d) Now the electric field is switched off and two different particles with masses m_1 and m_2 and same charge are accelerated. Describe the trajectories of the particles in MS. Determine the mass difference between the two particles if a particle detector placed along the z-axis records hits of two particles at positions and $-z_1$ and $-z_2$ respectively.
Hinweis: This part can be solved without knowing the solutions from b) and c).

Exercise 4: Cyclotron

Objective: Lorentz force, circular motion, (nonrelativistic) kinetic energy, cyclotron frequency.

Consider some protons being accelerated inside a cyclotron. Their trajectory is guided by an homogeneous magnetic field $\vec{B}$, while they pass through the two (D-shaped) semicircles sketched in the figure. A high frequency alternating voltage V is applied between the two semicircles, in such a way that an electric field appears in the gap between the semicircles. We consider the electric field to disappear inside the semicircles (acting as Faraday cages). A proton source Q is positioned near the middle, emitting protons with a low initial velocity. Before the protons leaves the cyclotron, their trajectory reaches the diameter D. Only the nonrelativistic case shall be considered and any higher harmonics of the fundamental frequency for the applied alternating voltage should be neglected.



a) What is the kinetic energy of the protons leaving the cyclotron?

b) What is the required frequency f of the alternating voltage?

c) What is the number N of turns, the protons make inside the cyclotron, and what time t_0 does it take for a proton to leave the cyclotron?

Exercise 5: Biot-Savart "around the corner"

Objective: Application of the law of Biot-Savart.

The Biot-Savart law can be used to calculate the magnetic field $\vec{B}$ of current-carrying components. As an example, the magnetic fields $\vec{B}$ of an infinitely long wire and of the inside of a conductor loop are calculated in the script. In this task, we want to use Biot-Savart's law to determine the magnetic field in the center of a "curve" (see sketch). The curve has a radius R and both ends of the curve extend to infinity.

a) Write the $\vec{B}$-field at the origin as an integral over the current and the tangent vector to the wire.

b) Decompose the expression into a sum of three components from which the magnetic field for the two straight sections and the curved section can be determined.

c) Calculate the three integrals.

d) Finally, determine the total magnetic field $\vec{B}$ in the center of the curve as a function of the current strength I and the radius R. Does the result meet your expectations?